LEARNING RESOURCE GUIDE

Generative AI for Beginners | Professional Participant Workbook

Section	Section 7: Prompt Engineering Mastery
Lecture	Lecture 7.4: Prompt Engineering for Different Modalities

Learning Objectives

By the end of this lecture and its accompanying guide, you will be able to:

- Distinguish the prompting requirements for text, image, audio, and video generation modalities.
- Apply the subject-action-setting-style framework to construct effective image generation prompts.
- Identify the unique prompt dimensions relevant to audio and music generation systems.
- Construct multimodal prompts that coordinate across text and visual generation for a professional deliverable.
- Evaluate prompt strategies for consistency and coherence across a multi-modal content project.

Introduction

Generative AI now spans multiple modalities—text, images, audio, video, and combinations thereof. While the core principle of prompt engineering applies across all modalities (specificity and context drive quality), the specific dimensions that matter differ significantly between them. A text prompt succeeds by specifying audience, tone, and format; an image prompt succeeds by specifying visual composition, style, and lighting. Practitioners who understand these differences can leverage the full range of generative AI tools rather than remaining confined to text generation alone.

The professional relevance of multi-modal prompting is growing rapidly. Marketing teams produce brand assets using text-to-image tools; training departments use text-to-speech synthesis for e-learning content; product teams prototype interfaces using AI-generated visuals. Each use case requires a distinct prompting vocabulary and evaluation framework. The principles learned in text prompting transfer partially but not completely—image generation, for example, is typically one-shot rather than conversational, placing a premium on comprehensive upfront prompts.

This guide provides a structured framework for prompting across modalities, with practical templates and professional examples for each modality type, and guidance for multi-modal workflows that coordinate generation across modalities—increasingly common in professional AI use.

Core Concepts Explained

Image Generation Prompting Dimensions

Definition: Image prompts succeed when they explicitly address seven dimensions: Subject (main focus), Details (specific attributes), Setting (environment), Style (artistic approach), Lighting (mood and atmosphere), Composition (framing and perspective), and Quality modifiers (resolution, detail level).

Business relevance: Professional image generation for brand assets, marketing materials, or product illustrations requires precise control over visual identity. The seven-dimension framework enables practitioners without design backgrounds to produce images that meet brand standards and communicate specific visual concepts.

AI Practitioner relevance: AI practitioners working with marketing, design, or communications teams use the seven-dimension framework to translate written briefs into effective image prompts. This skill is increasingly listed in job descriptions for AI product managers, content strategists, and digital marketing specialists.

Real-world example: Coca-Cola's 2023 AI-generated advertising campaign used highly engineered prompts specifying brand colours ('deep red, white, and silver'), product positioning ('centred, 30-degree angle'), setting ('festive table with winter decorations'), and style ('photorealistic, warm golden lighting') to align outputs with brand guidelines without extensive post-production editing.

Risks or limitations: Image generation models can produce outputs that technically satisfy all prompt dimensions but violate brand guidelines, cultural expectations, or copyright norms in subtle ways. Professional image generation workflows should include human review for brand and appropriateness checks before any output is published or used commercially.

Audio and Voice Generation Prompting

Definition: Audio generation prompts specify acoustic dimensions: voice characteristics (gender, age, accent), emotional tone, speaking pace, pauses, and for music—genre, tempo, instrumentation, mood, and structure.

Business relevance: Text-to-speech synthesis is now used for e-learning narration, customer service bots, podcast production, and accessibility tools. Effective audio prompts reduce post-production editing and ensure outputs match the intended audience and use case.

AI Practitioner relevance: Practitioners building AI-powered learning content or customer experience tools need to understand audio prompting to spec voice synthesis effectively. Specifying the wrong tone or pace can undermine the learner experience or customer perception regardless of the text content quality.

Real-world example: Duolingo's AI-generated audio lessons use prompts that specify not just spoken content but speaking pace ('deliberate, clear enunciation'), tone ('encouraging and patient'), and pronunciation emphasis to produce audio that supports language learners rather than native speakers.

Risks or limitations: AI-generated voices can inadvertently convey unintended emotional tone or exhibit uncanny characteristics that undermine listener trust. Professional audio generation should include audience testing before deployment, particularly for sensitive contexts such as healthcare or financial services.

Video Generation Prompting

Definition: Video generation prompts extend image prompting with temporal dimensions: motion description (what moves and how), shot sequence, duration, and narrative arc within the clip.

Business relevance: AI video generation is used for explainer content, product demonstrations, social media assets, and training materials. Effective video prompts specify not just visual content but visual storytelling—what changes over time and why.

AI Practitioner relevance: Practitioners producing AI video for marketing or training contexts must learn to think in shots rather than static compositions. Concepts from video production—establishing shots, close-ups, camera movement—become part of the prompting vocabulary.

Real-world example: Runway ML users producing product launch videos use prompts that specify motion ('camera slowly pushes in towards the product', 'product rotates 360 degrees on a white background') combined with lighting and style specifications to produce usable marketing video without a full production crew.

Risks or limitations: Video generation models in 2025 remain prone to temporal inconsistencies—objects that appear and disappear between frames, unnatural motion. Professional video generation should include frame-by-frame review for any output featuring human subjects or brand assets.

Multi-Modal Prompt Coordination

Definition: Multi-modal workflows coordinate prompts across two or more generation modalities to produce a coherent deliverable—for example, generating both narration script and accompanying visuals for an e-learning module from a single brief.

Business relevance: Coordination reduces inconsistency between modalities: image style matches text tone; audio pacing matches visual complexity; the overall deliverable has a unified register. Without coordination, multi-modal AI outputs often feel assembled from unrelated components.

AI Practitioner relevance: Practitioners leading multi-modal AI projects should author a shared prompt brief that defines cross-modal consistency standards before any generation begins. This brief functions as a style guide for the AI workflow.

Real-world example: An IBM design team producing AI-assisted product documentation created a shared prompt brief specifying text tone ('professional, approachable, second-person'), image style ('clean line illustrations, IBM blue palette'), and audio voice ('measured pace, neutral North American accent'). All three modalities were prompted against this brief, producing a cohesive product requiring minimal editorial reconciliation.

Risks or limitations: Coordination adds planning overhead that single-modality projects do not require. Practitioners should invest this time upfront—a shared brief takes an hour to write and saves many hours of post-hoc reconciliation across modalities.

Practical Applications

The following professional scenarios illustrate multi-modal prompting in practice.

E-Learning Module Production

A learning and development team produces a 10-minute compliance training module using AI across three modalities. Text prompting generates the narration script (tone: professional, audience: non-specialist employees, 60-word paragraphs). Image prompting generates scenario illustrations (style: flat vector, warm corporate palette, diverse professional environment). Audio prompting generates narration (voice: measured, clear, gender-neutral, 130 words per minute). All three prompts reference a shared consistency brief: professional tone, diversity representation, compliance-appropriate gravity. The result requires one revision cycle rather than four.

Brand Asset Generation

A startup's marketing team uses AI to generate initial brand assets for a pitch deck. The prompt brief specifies: industry ('clean energy'), brand personality ('innovative but grounded, trustworthy'), colour palette ('forest green, slate grey, white'), and image style ('photorealistic, natural environments with human presence'). Prompts for each individual asset reference the brief. Outputs are reviewed by a designer for brand coherence rather than created from scratch, reducing design time by approximately 60%.

Product Demonstration Video

A software product team needs a 60-second demo video. Using video generation prompting: Shot 1 prompt specifies 'UI screen recording style, clean white background, cursor navigates to feature menu, smooth motion, 8 seconds.' Shot 2: 'Split-screen: left shows before state, right shows after state, zoom in on the key metric that changed, 10 seconds.' A text-to-speech narration with matching timing specifications is assembled separately. The result is a functional demo for social media requiring no video production crew.

Best Practices

- **Use modality-specific vocabularies.** Image prompts benefit from art direction terminology (chiaroscuro lighting, shallow depth of field). Audio prompts benefit from audio production terms (reverb, dynamic range, room tone). Learning modality vocabulary significantly improves prompt precision.
- **Front-load comprehensive prompts for one-shot modalities.** Image and video generation is typically one-shot—you cannot have a dialogue to refine outputs as you can with text. Invest more time in the initial prompt; include all key dimensions before generating.
- **Create a shared prompt brief for multi-modal projects.** Define cross-modal consistency standards (tone, style, palette, voice) in a shared document before authoring individual prompts. This prevents the 'assembled from different sources' aesthetic that characterises poorly coordinated multi-modal outputs.
- **Evaluate outputs against the brief, not just the prompt.** Each generated output should be evaluated against the overall project brief. A visually striking image that violates brand guidelines fails the brief even if it technically satisfies the prompt.
- **Build in human review for brand, cultural, and legal checks.** Automated generation can produce outputs that violate brand guidelines, contain culturally inappropriate content, or raise copyright concerns. Professional workflows should include human review before any AI-generated asset is published or shared externally.

Common Mistakes and Misconceptions

Misconception: Treating image prompts like text prompts—using abstract descriptions rather than visual specifications.

Correct approach: Image prompts should describe what the image literally looks like: subject, setting, composition, style, lighting. Abstract concepts like 'innovative' must be translated into visual equivalents: 'clean geometric design, forward-leaning subject, modern office setting.'

Misconception: Neglecting temporal dimensions in video prompts.

Correct approach: Video is not a static image—it has motion, sequence, and narrative arc. Video prompts should specify what changes over time: camera movement direction and speed, subject motion, transitions between scenes.

Misconception: Generating across modalities without a shared consistency brief.

Correct approach: Without a brief, text, image, and audio outputs are generated in isolation and often feel tonally and stylistically inconsistent when assembled. Write the shared brief first; prompt all modalities against it.

Misconception: Assuming professional-quality assets can be published directly from generation without review.

Correct approach: AI-generated assets require professional review for brand alignment, cultural sensitivity, and copyright compliance. Generation speeds up production; it does not eliminate the need for human quality assurance.

Prompt Template: Multi-Modal Prompt Brief Template

Use this template at the start of any project generating content across more than one modality. Complete all rows before authoring individual prompts.

Parameter	Text Generation	Image Generation	Audio/Video Generation
Audience	Who will read this?	Who will view this?	Who will hear/watch this?
Tone / Style	Formal / conversational / technical	Photorealistic / illustrative / abstract	Professional / warm / authoritative
Subject / Topic	What is the text about?	What is the image's main subject?	What does the audio/video depict?
Format / Composition	Essay / list / dialogue	Close-up / wide / overhead shot	Duration / shot sequence / pacing
Quality / Length	Word count / sections	Resolution / detail level	Pace (WPM or BPM)
Brand / Consistency	Vocabulary, style guide	Colour palette, logo rules	Voice, music bed specs

How to use this: *Complete this brief before any generation begins. Share it with all team members prompting in different modalities to ensure cross-modal consistency.*

Knowledge Check

Test your understanding before moving on. Answers are shown below each question.

1. A practitioner generating an image for a financial services brand specifies: 'A photo about trust and stability.' The output is inconsistent with the brand's visual identity. What is the most likely cause?

A. The image generation model cannot process abstract brand concepts.

B. The prompt uses abstract concepts rather than concrete visual specifications such as composition, setting, and style.

C. Financial services imagery requires a specialised model.

D. The prompt is too short for image generation models.

Answer: B. The prompt uses abstract concepts rather than concrete visual specifications such as composition, setting, and style.

Why: *Image generation requires visual specifications—what the image literally shows—rather than abstract brand concepts. 'Trust' must be translated into visual elements: subjects, expressions, settings, colours.*

2. A team producing a multi-modal compliance training module using AI-generated text, illustrations, and audio should create what before authoring any individual prompts?

A. A complete draft of the text content for the AI to adapt for other modalities.

B. A shared prompt brief defining tone, style, and consistency standards across all three modalities.

C. A set of few-shot examples for each modality individually.

D. A chain-of-thought prompt that generates all modalities in a single pass.

Answer: B. A shared prompt brief defining tone, style, and consistency standards across all three modalities.

Why: *A shared prompt brief ensures cross-modal consistency—tone, visual style, audio register—preventing the assembled output from feeling like unrelated components.*

3. Why is front-loading comprehensive detail in the initial image prompt more important than in a text conversation prompt?

A. Image generation models are less capable than text models and need more instruction.

B. Image generation is typically one-shot—there is no conversational dialogue to refine the output iteratively.

C. Image prompts are limited in length, so efficiency is essential.

D. Image generation models ignore any context added after the first sentence.

Answer: B. Image generation is typically one-shot—there is no conversational dialogue to refine the output iteratively.

Why: *Unlike text generation where dialogue allows iterative refinement, image generation produces an output from a single prompt; all key dimensions must be specified upfront.*

Reflection Questions

1. Consider a professional deliverable you currently produce. Which modalities could AI assist with, and what would a shared prompt brief for that deliverable look like?
2. What new vocabulary would you need to learn to prompt effectively in a modality you currently do not use? Where would you start?
3. How would you evaluate an AI-generated image asset against a brand brief? What criteria would you use, and who in your team would be involved?
4. Think about a multi-modal project assembled inconsistently in your professional experience. How could a shared prompt brief have prevented those inconsistencies?
5. As video generation quality improves, what professional use cases in your industry are likely to be most disrupted? How should practitioners prepare?

Key Takeaways

- Each generative AI modality—text, image, audio, video—has distinct prompting dimensions; skills transfer partially but require modality-specific vocabulary and frameworks.
- Image prompts should specify seven concrete visual dimensions: subject, details, setting, style, lighting, composition, and quality modifiers—not abstract concepts.
- Image and video generation is typically one-shot; invest in comprehensive upfront prompts because iterative dialogue is not available to refine outputs.
- Audio prompts should specify voice characteristics, emotional tone, pace, and for music—genre, tempo, instrumentation, and mood.
- Multi-modal projects require a shared prompt brief defining cross-modal consistency standards before any individual prompts are authored.
- AI-generated assets across all modalities require professional review for brand alignment, cultural appropriateness, and copyright compliance before external use.
- The ability to prompt effectively across modalities is an increasingly valued professional skill as generative AI tools diversify beyond text generation.

Recommended Next Actions

6. Select one image generation tool and practice the seven-dimension framework on a professional image brief. Compare outputs with and without each dimension specified.
7. Draft a shared prompt brief for a multi-modal project in your professional context. Identify which cross-modal consistency parameters matter most for your use case.
8. Explore one audio generation tool and experiment with pace, tone, and voice specification. Evaluate how each parameter change affects suitability for a professional context.
9. Review your team's current AI usage. Which modalities are underutilised? Identify one new modality to experiment with in a low-stakes internal project.